Object class ==> It is parent class of any java class
It has 7 methods!
Concrete class, Concrtee method
Abstarct class, Abstarct method
// Both decalration + body ==> complete method ==> concrete method
clas sis having only conccrete methods ==> concrete class
method with only decalration but no body ==> abstartc methods
class concrete methods ==> concrete class
class both abtsract methods + concrete methods ==> abstarct class
if a class is fully implemneted(complete) ==> Concerete class
if a class is partially implemneted(complete) ==> Abstarct class
If a class is having any abstarct methods, for those methods body will be given by another java class(child class)

```
package com.abc.demo;

abstract class Vehicle
{
   public void engine()
   {
      System.out.println("Every vehcile has a engine!");
   }

   abstract public void wheels();
}

class Car extends Vehicle
{
   public void wheels()
   {
      System.out.println("This vehcile has 4 wheels");
   }
}

public class Third {

   public static void main(String[] args) {
      // TODO Auto-generated method stub

   }

}
```

1990's ==> Java was indroduced
before 1990's ==> c, c++
1990's ==> Databases ==> oracle, db2, sybase,.....
Java was introcuted ==> James Gosling
companies ==> java ==> develop any app
application

---
oracle
sybase
db2

```
connect(){
----------
}
disconenct()
{
}
```

```
connection(){
----------
}
disconenction()
{
}
```

```
con(){
----------
}
discon()
{
}
```

---
document

```
connect()
insert()
```

---
abstarct methods
INTERFACE ==> It is a speccification / document
that contains only abstract methods!

```
package com.abc.demo;


interface Database
{
   public void connect();
   public void insert();
   public void read();
   public void disconnect();
}

class Oracle implements Database
{
   public void connect()
   {
      System.out.println("connection logic.....");
   }

   public void insert()
   {
      System.out.println("insertion logic.....");
   }

   public void read()
   {
      System.out.println("read logic.....");
   }

   public void disconnect()
   {
      System.out.println("disconnection logic.....");
   }
}

class Sybase
{

}

class DB2
{

}


public class Fourth {

   public static void main(String[] args) {
      // TODO Auto-generated method stub

   }

}
package com.abc.demo;

interface camera
{
   public void capturePhoto();
}

interface musicplayer
{
   public void playMusic();
}

class SmartPhone implements camera, musicplayer
{
   public void capturePhoto()
   {
      System.out.println("capturing photo.....");
   }

   public void playMusic()
   {
      System.out.println("playing music.....");
   }
}


public class Fifth {
   public static void main(String[] args) {

   }
}
```

Concerete class ==> zero shareability
Abstarct class ==> partial shareability
Interface ==> full shareability
single inheritance ==> one child and for that one child, one parent

```
class AA
{

}

class BB extends AA
{

}
```

multilevel inherticance ==> inheriatnce that exists at multiple layers

```
class AA
{

}

class BB extends AA
{

}

class CC extends BB
{

}

class DD extends CC
{

}
```

Heirarchial Inheritance ==> multiple chil same parent

```
class AA
{

}

class BB extends AA
{

}
class CC extends AA
{
}
```

Mulriplate Inheritance ==> multiple parent for single child
It is not possible in JAVA using classes but it is posisble using interfaces

```
package com.abc.demo;

class AA
{
   public void display()
   {
      System.out.println("hello");
   }
}

class BB
{
   public void display()
   {
      System.out.println("welcome");
   }
}

class CC extends AA, BB
{
//   2 methods ==> display(), display()
}



public class Sixth {

   public static void main(String[] args) {

      CC c = new CC();
      c.display();

   }

}
```

Multiple Inheritance with Interfaces

```
package com.abc.demo;

interface camera
{
   public void capturePhoto();
}

interface musicplayer
{
   public void playMusic();
}

class SmartPhone implements camera, musicplayer
{
   public void capturePhoto()
   {
      System.out.println("capturing photo.....");
   }

   public void playMusic()
   {
      System.out.println("playing music.....");
   }
}


public class Fifth {
   public static void main(String[] args) {

   }
}
```